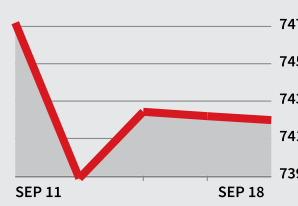


the hindu businessline

SENSEX 74294.96 (-486.80)



IN FOCUS

	Week's close	Week's change
Nifty 50	23346.40	-51.70
P/E Ratio (Sensex)	19.68	-0.13
US Dollar (in ₹)	95.87	+0.32
Gold Std 10 gm (in ₹)	153111.00	+1781
Silver 1 kg (in ₹)	236908.00	+7988

QUANTUM.

Priyansh Mohan's start-up Meine Electric develops iron-air battery system that offers long-duration renewable energy storage **p7**



PULSE.

Medical bodies can now access patient data in exceptional cases, but with caveats **p9**

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QUICKLY.

STRONG PIPELINE

72 firms await SEBI nod to raise ₹1.7 lakh cr via IPOs



Mumbai: The pipeline of initial public offerings (IPOs) is expected to remain robust in the coming months, with 72 companies waiting for SEBI approval to raise about ₹1.7 lakh crore as of August-end, according to data sourced from primedatabase.com. Despite bearish sentiment in the secondary market, it has already been raining IPOs, with 20 companies raising about ₹12,269 crore so far in September. **p2**

TO FIGHT DRONES

DRDO's directed-energy weapon to be ready soon

New Delhi: The Defence Research and Development Organisation's (DRDO) first-of-production model (FOPM) for the 30 kW-class directed-energy weapon (DEW), a strong arm of the 'Sudarshan Chakra' comprehensive air defence system announced by Prime Minister Narendra Modi earlier, will be available within a year to counter swarm drones and unmanned aerial vehicles. **p10**

For India, border peace with China comes before trade ties

CAUTIOUS STANCE. Delhi holding back on big concessions until border stability improves

Amiti Sen
New Delhi



STRATEGIC RESTRAINT. A file photo of Commerce Minister Piyush Goyal with his Chinese counterpart Wang Wentao

India is not ready to offer big concessions to China on trade and investment until there is greater assurance on peace and stability along the border despite the recent high-level economic engagement between the countries during Chinese President Xi Jinping's visit for the BRICS Summit, sources said.

The position assumes significance following the meeting between Commerce and Industry Minister Piyush Goyal and his Chinese counterpart Wang Wentao on the sidelines of the BRICS Summit in New Delhi earlier this month to resolve long-pending concerns on both sides.

"While the meeting marked a resumption of high-level trade talks, India is not looking at moving ahead on market-access or investment concessions without progress on its concerns relating to the border," a source tracking the matter told *businessline*.

TRADE TALKS

Xi visited India this month after seven years as ties had

soured following the 2020 Galwan Valley clash, leading to prolonged tensions along the Line of Actual Control (LAC). The visit, on September 12, came amid renewed efforts by India and China to ease tensions, although recent disputes over alleged Chinese encroachment have reinforced India's priority of maintaining peace and stability along the border.

At the meeting between the two trade Ministers, India raised China's restrictions on exports of rare earth minerals and high-technology products, besides flagging the large and widening trade deficit.

"Beijing did not provide specific assurances on these issues," another official said.

While bilateral trade between India and China touched a record \$151 billion in FY26, the balance remains heavily skewed in China's favour, with imports from China into India at \$131.6 billion.

CHINA'S CONCERN

China, meanwhile, raised India's restrictions on investments from Chinese companies and sought greater liberalisation of the investment regime, the official added.

FDI from countries sharing a land border with India, including China, require prior government permission, which could take a long time.

"Although some conces-

sions were made for investments coming in from border countries, including China, earlier this year, it was nominal and Beijing wants the investment restrictions to be totally removed. But there is not much scope for anything substantial happening now till Beijing contributes to improving the border situation," the first source said.

At their bilateral meeting on the sidelines of the BRICS Summit, Prime Minister Narendra Modi told Chinese President Xi Jinping that "peace and tranquillity" in the border areas remained an "essential basis" for the continued development of bilateral relations.

He also stressed the need to observe existing agreements and understandings on border-related issues.

Xi, on the other hand, called for the two countries to take a strategic and long-term view of their relationship, cooperate for common development and advance bilateral relations alongside efforts to resolve the boundary question. China's readout called for "parallel, mutually reinforcing progress" on the two fronts.

MoSPI plans to release pre-FY23 GDP back series by year-end

Shishir Sinha
New Delhi

With the change in base year to 2022-23 from 2011-12, India plans to release back series GDP data for years prior to 2022-23 by the end of this year, said MoSPI Secretary Saurabh Garg.

"Releasing two official growths of GDP by running the old and new series concurrently would add to more ambiguity and confusion among the users and, therefore, none of the countries run the parallel series together," he told *businessline*.

Interview p3

SEBI board may discuss PMS, settlement norms

Akshata Gorde
Mumbai

The SEBI board, which is scheduled to meet on September 24, may consider an overhaul of settlement regulations and portfolio management services (PMS), a common advertisement code for regulated entities and allowing FPIs access to commodity derivatives.

Details p2

Tata Trusts cites SC's Cyrus Mistry ruling to counter Chandra

Our Bureau
Mumbai

The challenge mounted by the Tata Trusts to the re-appointment of N Chandrasekaran as Chairman of Tata Sons draws its strength from the Supreme Court's ruling in the Cyrus Mistry case, where the court upheld the sanctity of the Tata Sons' Articles of Association (AoA) and recognised the special rights accorded to Tata Trusts as the majority shareholder.

In a statement on Sunday, the Trusts argued that the September 17 board resolution granting Chandrasekaran a fresh five-year term failed to satisfy a mandatory requirement under the AoA.

According to the Trusts, certain decisions of the Tata Sons board require the affirmative support of a majority of Tata Trusts-nominated directors. With one of the two Trust nominees voting against the resolution, that condition was not met, rendering the resolution invalid regardless of the overall board vote.

The Trusts maintained that the Chairman's casting vote could not be used to overcome this deficiency. While a casting vote may

break a tie in the overall board vote, it could not override a separate affirmative voting requirement that is explicitly embedded in the company's constitution. "The Board put a question, and the AoA answered it in the negative," the Trusts said, arguing that the resolution was void *ab initio* (void from the outset).

LEGAL BATTLE

The Trusts' argument is closely tied to the position adopted by Tata Sons during the legal battle stemming from Mistry's removal as Chairman. In those proceedings, Articles 104B and 121, which confer affirmative voting rights on Tata Trusts-nominated directors, were challenged as oppressive.

Tata Sons strongly defended those provisions, arguing that they were legitimate protections available to the majority shareholder rather than instruments of oppression. The Supreme Court ultimately endorsed that view and upheld the validity of the Articles.

The Trusts now contend that Tata Sons cannot depart from an interpretation of the AoA that it successfully persuaded the country's highest court to uphold.

Also read p10

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QUICKLY.

BEML bags ₹5,400 cr order for 16 high-speed trains

Ahmedabad: BEML Ltd has secured a ₹5,400 crore contract from the National High Speed Rail Corporation Ltd to supply and maintain 16 high-speed trains for the Mumbai-Ahmedabad High Speed Rail corridor, marking the largest single order in the company's history and taking its overall order book beyond ₹21,000 crore. These train-sets have to be delivered in the next seven-eight years, official sources told *businessline*. **OUR BUREAU**

Mother Dairy sets up dedicated butter chiplet

Ahmedabad: Mother Dairy has set up its first dedicated butter chiplet manufacturing line in Junagadh, Gujarat, with an annual production of 550 tonnes, as the firm looks to expand its presence in the butter segment. The new line has been established at Mother Dairy's existing milk processing facility in Junagadh and will supply butter chiplets across India, a company official said on Saturday. **OUR BUREAU**

SEBI board to consider settlement, PMS, FPI rule changes on Sept 24

ON THE AGENDA. FPI participation in commodity derivatives, advertising framework may also be discussed

Akshata Gorde
Mumbai

The SEBI board is scheduled to meet on September 24 and is expected consider an overhaul of settlement regulations and portfolio management services (PMS), a common advertisement code for regulated entities and greater access for foreign portfolio investors (FPIs) to commodity derivatives, according to people familiar with the matter.

The board may also take up proposals relating to depository receipts for REITs and InvITs, expansion of the regulatory framework for vault managers and changes to debt market rules, among others.

MF PORTFOLIOS

The markets regulator has proposed a comprehensive review of regulations governing portfolio managers,

KEY PROPOSALS

- Settlement regulations may see a comprehensive overhaul
- PMS regulatory framework could be revamped
- Wider FPI participation in non-agri commodity derivatives proposed
- Common ad code, REIT/InvIT DRs and vault manager changes may also be considered

including allowing mutual-fund-only PMS wherein portfolio managers could offer professionally managed portfolios comprising mutual fund schemes.

It has also suggested widening the investment universe available to PMS firms, including permitting investments in yet-to-be-listed securities and allowing discretionary portfolio managers to invest up to 10 per cent of a client's assets under

management in investment-grade unlisted debt securities.

SEBI has also proposed an overhaul of the settlement framework, including changes aimed at rationalising settlement amounts and speeding up smaller cases.

The proposals also seek to allow settlement at later stages of proceedings, including when a matter is before the Securities Appellate

Tribunal or the Supreme Court, and introduce a fast-track route for specified violations and cases involving settlement amounts of up to ₹10 lakh.

Other changes include issuing settlement notices before show-cause notices in certain cases, extending the time to apply for settlement after a show-cause notice from 60 days to 90 days and reducing the additional amount payable for refiling a withdrawn settlement application from 50 per cent to 20 per cent.

FPI EXPANSION

Another proposal seeks to widen FPI participation in exchange-traded commodity derivatives. SEBI has proposed allowing FPIs to participate in non-agricultural commodity index derivatives and in non-agricultural commodity derivative contracts that are settled through physical delivery,

subject to prescribed conditions.

The board may also consider SEBI's proposed common advertisement code for regulated entities, which seeks to bring advertising requirements for different regulated entities under a common framework and includes provisions allowing celebrity endorsements by regulated entities subject to conditions.

The regulator has also proposed expanding the scope of its Vault Managers Regulations, 2021, which currently govern vault managers providing services for electronic gold receipts traded on recognised stock exchanges.

Separately, SEBI has proposed enabling depository receipts against units of REITs and publicly listed InvITs, which will allow these units to be accessed by investors through overseas markets.

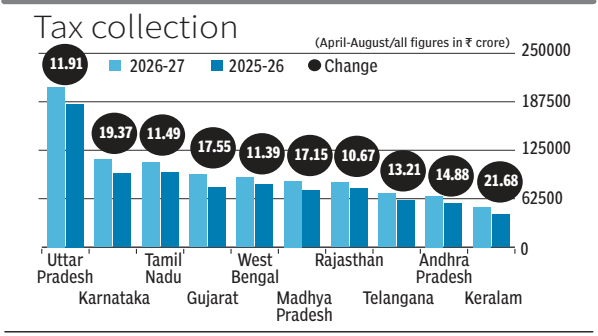
U.P. affirms top position in tax collection, followed by Karnataka, T.N.

Shishir Sinha
New Delhi

Uttar Pradesh continues to lead the States and Union Territories in collection from taxes during the April-August period of the current fiscal, followed by Karnataka and Tamil Nadu, data from the Comptroller & Auditor General (CAG) show.

The State also tops in terms of the highest number of GST assesses with over 22 lakh. Uttar Pradesh collected over ₹2.06 lakh crore during the five-month period, is around 12 per cent higher than the corresponding period of the last fiscal.

Keralam, though last in absolute numbers among 10 States (*see table*), recorded



the highest growth at around 22 per cent. The study does not include Maharashtra, as five-month data for the said State is not available.

GST GROWTH

The States' tax revenue comprises seven components — State Goods & Services Tax

(SGST), stamps & registration, land revenue, sales tax, State excise duty, the State's share of union taxes and other taxes & duties.

Among these, SGST has the maximum share. All the 10 States taken for this report have shown collection from GST increasing despite

rate rationalisation and doing away with compensation. For example, according to data compiled by the CAG, Uttar Pradesh earned over ₹73,000 crore through SGST during the April-July period of the current fiscal as against over ₹66,000 crore during the corresponding period of last fiscal.

In terms of capital expenditure, Gujarat was ahead of the others with spending of over ₹29,600 crore in the first five months of the current fiscal as against around ₹26,000 crore during the corresponding period.

At the same time, while Uttar Pradesh spent over ₹21,000 crore during the five-month period, like the corresponding period of the

last fiscal, the amount for Karnataka was over ₹15,600 crore against around ₹14,000 crore.

All these data are being studied at a time when the Centre has convened a two-day conference over the weekend with the States and the Union Territories (with legislatures) on "Financing India's Journey towards Viksit Bharat."

CONFERENCE TALK

During the conference, while Economic Affairs Secretary Anuradha Thakur emphasised the need for higher private finance, Chairman of the 15th Finance Commission NK Singh expanded the deliberations by suggesting measures for revenue augmentation by the States, be-

sides issues such as debt-GDP ratio and going from public finance to private capital, among others.

According to Thakur, keeping in mind the ongoing global difficulties and because of the scale of transformation, "We envisage ourselves to reach our shared goal. It needs to be borne in mind that this scale cannot be met solely by government budgets, and that private sector financing will need to play a critical role."

Suggesting ways for how the States could boost revenue, he said they were better placed to leverage their local information advantage. The opportunity, therefore, is to turn this information advantage into stronger revenue mobilisation.

Adani Power lays foundation stone for ₹48,000 crore thermal project in Assam

Our Bureau
Ahmedabad

The foundation stone for Adani Power's 3,200 MW thermal power project was laid in Assam on Sunday, marking an investment of around ₹48,000 crore and setting the stage for what the company says will be the "North-East's largest power generation asset".

The project is being developed at Chapar in Assam's Dhubri district. Speaking at the foundation stone ceremony, Jeet Adani, Director, Adani Group, said the company had already conducted four to five job drives in Assam, with around 4,500 local youth hired for the project that will bring together thousands of workers, engineers, technicians, contractors and suppliers.

The 3,200 MW project



Jeet Adani, Director, Adani Group

marks a significant expansion of Adani Power's presence in Assam, and Adani said the group's businesses in Assam are spread across sectors including power, airports, smart meters, cement, connectivity and maintenance, repair and overhaul services.

'BATTERY OF INDIA'

"Assam has the potential to become the 'Battery of India,'" Adani said, outlining

the group's long-term view of the State's energy potential.

The foundation stone ceremony comes against the backdrop of the Adani Group's broader investment plans for the North-East.

At the Rising Northeast Global Investors Summit last year, Group Chairman Gautam Adani had announced plans to invest up to ₹1 lakh crore in the North-East over the next decade.

Jeet Adani said work was already underway in Assam with investments of nearly ₹80,000 crore.

Beyond power generation, the Group is also developing and operating Guwahati's Lokpriya Gopinath Bordoloi International Airport, where it plans to develop an airport city with commercial, hospitality, retail and entertainment infrastructure.

Stellantis takes full ownership of Tiruvallur plant, buys out stake of CK Birla Group

T E Raja Simhan
Chennai

Stellantis India has taken full ownership of its Tiruvallur plant by buying the stake of Hindustan Motor Finance Corporation Ltd (HMFCL), a CK Birla Group company, in the joint venture entity Stellantis Automobiles India Pvt Ltd (SAIPL).

The details of the amount invested in the transaction and the percentage of stake bought were not disclosed.

The acquisition would provide greater flexibility and agility in decision-making and enable the company to respond faster to market and customer requirements, Shailesh Hazela, CEO and Managing Director, Stellantis India, told *businessline*.

"Both partners agreed to keep the information on

equity sold and amount paid confidential," he added. The firm is working with the Tamil Nadu government for additional investment linked to the production ramp-up and it will be announced, Hazela said.

Stellantis and the CK Birla Group began their manufacturing and mobility partnership in 2017. The Tiruvallur facility currently manufactures the Citroën C3, ë-C3, C3 Aircross and Basalt. The plant has achieved more than 95 per cent localisation, per Hazela.

"Stellantis has invested over €1 billion in India to support its long-term growth ambitions, including manufacturing, product development, localisation and capability building. The Tiruvallur facility benefits from a strong supplier and logistics ecosystem while



Shailesh Hazela, CEO and MD, Stellantis India

contributing to Tamil Nadu's position as a leading automotive manufacturing hub," he said.

FUTURE LOOKOUT

Stellantis is targeting to produce over 50,000 units annually by 2027 from 18,000 units in 2026, which is also expected to increase employment at the facility. Cur-

72 companies await SEBI nod to raise ₹1.7 lakh cr via IPOs

Suresh P Iyengar
Mumbai

Waiting for SEBI approval

	Issue size (₹ cr)
Cult.Fit	3,500
Tablespace Technologies	3,350
Svatantra Microfin	3,000
Muthoot Fincorp	3,000
Encube Ethicals	3,000
Veritas Finance	2,800
Sagar Defence Engineering	2,000
Atomberg Technologies	1,500
Assetz	1,200

Source: primedatabase.com

The pipeline of initial public offerings (IPOs) is expected to remain robust in the coming months, with 72 companies waiting for SEBI approval to raise about ₹1.7 lakh crore as of August-end, according to data sourced from primedatabase.com.

Some of the companies awaiting SEBI approval for DRHPs include Mahanadi Coalfields (₹10,000 crore), Carlsberg India (₹6,300 crore), Sembcorp Green Infra (₹3,750 crore), Emerald Jewel Industry India (₹3,500 crore), Encube Ethicals (₹3,000 crore) and Veritas Finance (₹2,800 crore), it added.

Despite bearish sentiment in the secondary market, it has been raining IPOs, with 20 companies raising about ₹12,269 crore so far in September.

GATHERING PACE

In fact, a record six companies listed on a single day last week after attracting demand of ₹1.4 lakh crore against an issue size of ₹7,100 crore. The rush of IPOs is expected to gather further pace this month, with SEBI extending the validity of approvals for IPOs and rights issues expiring between April 1 and September 30, 2026, to September 30, 2026.

VK Vijayakumar, Chief Investment Strategist, Geojit Investments, said there is an element of irrational exuberance in the booming IPO market as most retail investors are bidding for listing gains and, therefore, are not giving due importance to valuations and management quality. As the average returns from IPOs during the period from early June to September 7 were 22 per cent, retail investors have been chasing IPOs, he added.

THE REAL CHALLENGE

Sandeep Gupta, Director, Valmiki Leela Capital, said

the real test for IPOs in the pipeline will be the willingness of issuers to launch issues at valuations that the market could absorb, and this could prompt some companies to wait for more supportive conditions.

Several 2026 listings are trading below their issue prices, showing that listing gains can no longer be taken for granted. Retail investors are increasingly paying attention to post-listing performance, which is putting greater emphasis on valuation discipline, he added.

"Jio Platforms will be closely watched, given its proposed fresh issue with no offer for sale. A well-received issue at a valuation that investors find attractive will reinforce confidence across the IPO pipeline," he said.

Anoosha Soham Bathwal, CEO & Founder of Dhanvestor, said there had been immense liquidity and enthusiasm in the market, especially from retail investors looking for a stake in India's success story. However, some weak listing performances by recently listed companies had provided a reality check for investors, she added.

"Investors should evaluate the business model, earnings predictability, competitive edge and, especially, valuation relative to peers. Even an excellent company can become overvalued once all future growth is priced in," said Bathwal.

P4 poverty initiative gains ground in A.P.

G Naga Sridhar
Hyderabad

The flagship poverty alleviation programme of Andhra Pradesh, Public-Private-People Partnership (P4), is steadily expanding its reach.

According to the latest government data, Palnadu district has recorded 98.61 per cent integration of village/ward secretariats with Bangaru Kutumbams (BKs), placing it among the districts with the highest level of linkage.

Palnadu is followed closely by Parvathipuram Manyam at 98 per cent, Visakhapatnam at 98.37 per cent and Guntur at 98.32 per cent. Anakapalli recorded 97 per cent integration, followed by NTR district at 97.69 per cent, Tirupati at 97.36 per cent, Nellore at 96 per cent and Srikakulam at 96.43 per cent.

The data indicate that several districts have achieved integration levels of more than 95 per cent. At the lower end, YSR Kadapa recorded 72.22 per cent, while Sri Sathya Sai stood at 76 per cent, Dr BR Ambedkar Konaseema at 79 per cent, Alluri Sitharama Raju at 80 per cent and Anantapur at 82 per cent.

A brainchild of Chief Minister N Chandrababu Naidu, the programme aims to bridge the socio-economic divide by encouraging the wealthiest 10 per cent of the population to support the bottom 20 per cent of the poorest families.

Core tech, engineering services lift India's FY26 IT export growth despite software product slowdown

Sindhu Hariharan
Chennai

India's software exports growth is being led by traditional information technology (IT) services and product design and engineering services, while the software product development and BPO services are seeing a slowdown, according to the Reserve Bank of India's latest survey on computer software and IT-enabled services exports for FY26.

SERVICES SURGE

Total software services exports rose from \$204.7 billion in FY25 to \$221.4 billion in FY26, an increase of 8.2 per cent.

This was driven by a 12 per cent and an 11 per cent growth respectively, in IT

Engineering services power India's IT exports (in \$ billion)			
	FY 2025	FY 2026	Growth %
IT services	131.3	147.0	12.0
Software product development	6.8	6.4	-5.9
BPO services	55.8	56.0	0.4
Engineering services	10.8	12.0	11.1
Total software services	204.7	221.4	8.2

Source: RBI

services and engineering services. The share of IT services in total exports rose sharply from 64.1 per cent in FY25 to 66.4 per cent of total software exports in FY26, a trend that analysts attributed to rising GCCs making up for the revenue slowdown of traditional IT firms.

In contrast, the share of software product development fell from 3.3 per cent to 2.9 per cent, considering the maturity of AI. Within

engineering services, the segment called 'product design engineering' (mechanical electronics, excluding software) saw a sharp rise in its share from 2.7 per cent to 4.3 per cent, suggesting that India is maturing as a hardware product engineering hub.

Further, embedded solutions' exports grew in share from 2.7 per cent to 3 per cent. This refers to the export of specialised engineering, design and software

services towards building embedded systems.

SOFTWARE EXPORT

Software exports to the US increased from \$108.3 billion in FY25 to \$119.7 billion, with the share of exports to the US rising from 52.9 per cent to 54.1 per cent.

In contrast, Europe's share slipped by one percentage point, despite exports to the region rising in absolute terms.

The RBI's survey also showed that the US dollar was the principal invoicing currency for India's software exports in the fiscal year, followed by the euro, the rupee, and the pound sterling.

In India, software services exports are primarily delivered through the off-site mode, which holds a 91.7 per cent share in FY26.

Grahaa Space accelerates launch plans for Earth-observation satellites

Aishwarya Kumar
Bengaluru

Bengaluru-based Grahaa Space is stepping up its launch cadence after successfully putting its SOL-ARAS satellite into orbit aboard Skyroot's Vikram-1, targeting two technology-demonstration missions in Q4 FY26 and Q1 FY27 and potentially three launches by the second half of FY27.

The start-up has already signed two customers for its upcoming missions, Co-founder and CEO Ramesh Kumar said in an interview. The missions will help Grahaa validate its optical payload, a key step before it begins building its planned constellation of nano satellites to deliver near-real-time video data of the Earth.

"The next mission is proving our optical payload. It is

the second technology demonstration mission," Kumar said, adding that once the payload is proven, "we will start building the constellation. There we will sell the data."

COMPANY'S VISION

The company's target, is to complete a satellite within 12 weeks, he said.

Grahaa currently gener-

ates revenue through its satellite bus and platform, which can be configured for research, industrial or academic payloads, as well as hosted-payload missions.

Its longer-term ambition is to build a constellation and monetise the resulting Earth-observation data across applications, including defence, agriculture and environmental monitoring.

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Release of pre-FY23 GDP back series likely by year-end: MoSPI Secretary

DATA RESET. Saurabh Garg discusses the fine print of the base year revision of the GDP

bl.interview

Shishir Sinha
New Delhi

The government is planning to release back series for years prior to 2022-23 by the end of this year, said Saurabh Garg, Secretary in the Ministry of Statistics and Programme Implementation (MoSPI). He discussed the fine print of the base year revision of the GDP.

Edited excerpts:

The key issue in the GDP debate is the use of double deflation and the use of producer/input prices. What new datasets has MoSPI obtained that make these methods feasible? The transition to double deflation was not sudden. It was built on extensive experiments and detailed analysis using historical time-series data from Annual Survey of Industries (ASI) spanning the period 2011-12 to 2023-24, supply and use tables (SUT) and the WPI of the 2011-12 series under the guidance of the Advisory Committee on National Accounts Statistics (ACNAS). The stability of output PPI from 2022-23 onwards was assessed and MoSPI examined the feasibility of using output PPI in national accounts prior to its official launch.

The DPIIT initiated the exercise for compilation of the Output Producer Price Index and hence, the base revision of GDP was an opportune time to introduce double deflation. Since output PPI was not released by February 27, double deflation was carried out using available item-level WPI (2011-12 series). Later on, the release of output PPI in June 2026, following out-

put PPI-based double deflation, was implemented and updated estimates were released on August 31, 2026.

What are the sector-wise price indices and weights producing the aggregate deflator? Given the big divergence between the GDP deflator and WPI/CPI, can MoSPI provide a reconciliation showing why the GDP deflator behaves so differently?

In the new base year, the deflation strategy in various sectors has been volume extrapolation, single extrapolation and double deflation; use of item-specific price indices has been emphasised in the new base compared to aggregate price indicators applied at the sectoral level.

Since in the old series on applying single deflation (the output and input are deflated by the same deflator), the movement in IPD (Implicit Price Deflator, commonly called the GDP deflator) of the National Accounts Statistics was close to the WPI/CPI. However, in the new series, the approaches used in constant price estimation are more sophisticated and hence, do not have a one-to-one correspondence with output PPI and CPI.

Under the new national accounts methodology, manufacturing uses double deflation: Output and intermediate inputs are separately deflated. Therefore, manufacturing GVA's implicit price change need not equal manufacturing PPI/WPI inflation when the output and input prices vary differently, and this fact is addressed in double deflation.

Similarly, CPI reflects price changes of a specific basket curated based on the household consumption of goods and services, whereas



In this new series, several methodological improvements have taken place. However, the major factor is mainly the new data for estimating the unincorporated sector

SAURABH GARG,
Secretary, MoSPI



GDP by demand-side covers government consumption, investment, export and import too, for which altogether separate price deflators have been adopted.

The GDP deflator is economy-wide: It captures price changes across all domestically produced final goods and services. WPI, by contrast, is a wholesale-price index for goods, with no comprehensive coverage of services. MoSPI explicitly notes that the GDP deflator need not move in line with WPI/PPI/CPI because of differences in their coverage and use of item-level price indices.

Why was the new methodology not run in parallel with the old one before it became the basis for the official series?

Several data sources, such as ASUSE, PLFS and output PPI, apart from methodological changes such as segregation of GVA of multi-activity enterprises, have been introduced in an integrated fashion in the new series. Bifurcation of growth in GVA to the one due to double deflation and that due to new data sources, coverage and other methodological changes is not feasible. Further, releasing two official growths of GDP by running the old and

new series concurrently would add to more ambiguity and confusion among users and, therefore none of the countries run the parallel series together. India is planning to release the back series for years prior to 2022-23 by the end of this year.

Can MoSPI demonstrate that the improvement is not merely a change in methodology but also an improvement in the quality and availability of the underlying data?

In earlier base series (e.g., 2011-12), the GVA of the unincorporated sector relied heavily on the Enterprise Survey conducted in the base year and the quinquennial round of the Employment Unemployment Survey. In subsequent years, the benchmark/base year estimates were extrapolated using relevant proxy indicators, such as growth in sales tax for extrapolating the base year GVA of trade. MoSPI now conducts the Annual Survey of Unincorporated Sector Enterprises (ASUSE), alongside the Periodic Labour Force Survey for labour force estimation on a continuous basis. Thus, the use of proxy indicators is not required.

Then, PFMS for various States were not available in the previous base. GST,

which is one of the important administrative data sources for quarterly GDP estimation, was launched in 2017-18. Eventually, the GSTN database has strengthened substantially.

Also, reconciling production-side GVA and expenditure-side GDP often resulted in sizeable discrepancies due to asymmetric data availability. In the new series, the compilation of GDP by production and expenditure approaches has been integrated with the SUT framework.

From ₹86 lakh crore to ₹80 lakh crore is a massive downward revision in the 2026 nominal GDP, and that has been questioned. What do you say?

Our nominal GDP in the new series has been revised down by about 2.7 per cent in 2022-23, 3.5 per cent in 2023-24 and 3.8 per cent during 2024-25. Such changes in base revision are due to the incorporation of better data sources and improved methodology. In this new series, several methodological improvements have taken place. However, the major factor, taking the levels down, is mainly the new data for estimating the unincorporated sector. Results of ASUSE and PLFS data provide a more direct basis for measurement of unincorporated sector contribution to value added on a regular basis in the new base year.

However, a proper assessment of the sector-wise picture shows revisions have happened both upward and downward. Agriculture and allied activities GVA were revised up by about 3.8-5.9 per cent; financial services and real estate, professional services and ownership of dwellings were revised up by approximately 7.8-9 per cent over the comparable years.

OMCs' under recovery on LPG up 5% in Aug

Rishi Ranjan Kala
New Delhi

The under recovery on LPG of PSU oil marketing companies (OMCs) rose roughly 5 per cent on a monthly basis to more than ₹62,000 crore in August 2026 as retailers continue to sell the critical cooking fuel at below the market rates.

The under-recovery was around ₹721 per cylinder in June, which fell to roughly ₹500 a cylinder a month later. The under recovery per cylinder stood about ₹188 in August, which has risen to roughly ₹210 a cylinder in the current month.

The cumulative under recovery of PSU OMCs rose from roughly ₹51,000 crore in June to over ₹62,000 crore last month. PSU OMC losses ballooned after they sold LPG below international prices, which surged significantly after February 28, due to the West Asia conflict and the closure of the Strait of Hormuz (SoH).

Prior to the outbreak of the conflict, India used to import about 60 per cent of its LPG consumption, of which about 90 per cent transited through the SoH.



LPG imports rose to a six-month high of 1.3 million tonnes (mt) in August, growing by roughly 50 per cent month-on-month. However, the cargoes were down by almost 33 per cent on an annual basis.

A back of the envelope calculation shows that for \$1 million, India procured roughly 1.65 tonnes of LPG in August 2026, roughly the same rate during March 2026. During April, around 1.39 tonnes of LPG was procured for \$1 million.

SUPPLY DISRUPTIONS However, the quantity fell as supply disruptions in the SoH completely choked off LPG supply during May and June.

For instance, India paid \$1 million to procure around 1.17 tonnes and 1.18 tonnes

of LPG in May and June, respectively.

The US was India's top supplier last month. India imported about 0.62 mt LPG from the US in August, after bringing in 0.89 mt in July, OPEC data showed.

Washington, which was India's fifth largest LPG supplier till January 2026, jumped a spot to become the fourth largest, replacing Kuwait a month later.

The fresh conflict in West Asia (from February 28) propelled the North American country to become the top supplier for five consecutive months, beginning March. However, this diversification has come at a cost.

The US LPG is linked to Mont Belvieu prices (\$0.68 per gallon in August 2026, equivalent to around \$360 per tonne) which is generally benched lower than Saudi CR, but the longer voyages to India raise its delivered cost, Equirus Securities pointed out.

Meanwhile, constrained regional supply pushed Saudi Arabia's September propane contract price up by \$5 per tonne to \$625. Higher sourcing costs are expected to continue to accumulate with Indian OMCs, it noted.

TODAY'S PICK.

KPR Mill (₹1,141.30): BUY

Akhil Nallamuthu
bl. research bureau

The stock of KPR Mill has been in an uptrend since April. Back then, it found support at ₹800 and established an uptrend.

Recently though, it has been witnessing higher volatility, particularly since the second half of June. While the trend has not turned bearish, the direction of the movement was largely sideways. But now it has found a support at ₹1,100, where a key rising trendline coincides. The rebound on Friday indicates it might gain more strength going forward.

We see a potential for the stock to appreciate to ₹1,400 in

the medium-term. So, buy now at ₹1,141 and ₹1,100. Place the stop-loss at ₹1,000. When the stock rallies to ₹1,250 and ₹1,340, alter the stop-loss to ₹1,190 and ₹1,300, respectively. Book profits at ₹1,400.

Note: The recommendations are based on technical analysis. There is a risk of loss in trading

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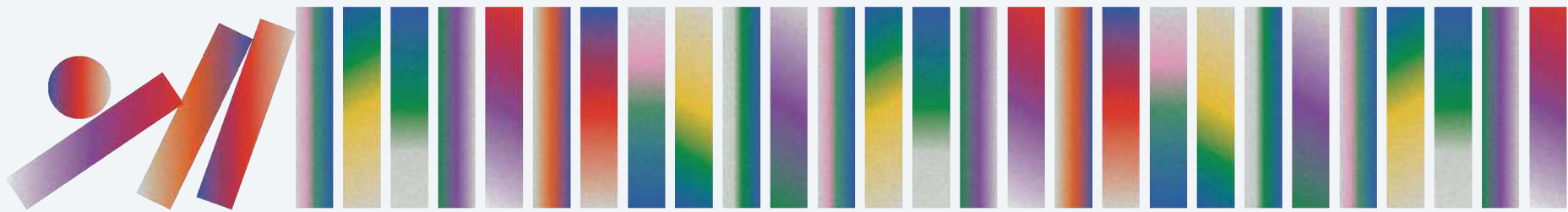
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Tale of a storied building

The history of the Parliament from within its walls

BOOK REVIEW.

TCA Srinivasa Raghavan

We are all used to hearing and reading gossip and stories about people. But about a building? Welcome to Sobhana K Nair's book about the old Parliament building, the huge round structure, whose design many believe was copied by Sir Edwin Lutyens from a 11th century temple in Morena near Gwalior. That temple is called Chausath Yogini Temple. The brief to the architects was to build something that looked Indian.

Fortunately, Nair hasn't dwelt on this aspect too much. She mentions it in passing because, after all, there's a much bigger tale to be told. An important element of that tale is that had it not been for Lutyen's obstinate insistence, the building would have been triangular. That's what Lutyen's partner, Herbert Baker, recommended and was overruled by the powers that were.

CIRCULAR OR TRIANGULAR? Baker then reluctantly fell in line and came up with the circular design. The triangular design was meant to represent the three major components of Indian polity: Council of State, Council of Princes and the Central Legislative Assembly.

He was, of course, most unhappy, but could do nothing about it. He and Lutyens had never got along much and Baker grumbled a lot about Lutyens. Nair quotes from his letters to his wife, in which he complains about Lutyens' nature.

He wasn't the only one complaining. Various people were just scornful and derisive about the round building. All sorts of nasty things were said about it. All in all, the *sahibs* were not happy, while the natives probably had no idea what the building was for.

These little titbits, from the history of the most important building in our country, are quite riveting. Nair also deals with the Central Hall issue.

The new Parliament building which, as it happens, finally has a triangular shape, but doesn't have a Central Hall. This has made MPs very sad because it was there that they met informally, exchanged gossip and struck deals.

Instead of that hall, there is a dining room now in the manner of British clubs, only much bigger. The new building doesn't seem to



Title: Housing the Republic
Author: Sobhana K Nair
Publisher: Bloomsbury Publishing India Pvt Ltd

have enthused Nair very much. She says it lacks soul.

Nair informs us that for 11 years after Independence in 1947, Parliament and the Supreme Court shared space in the building, with the latter occupying what used to be the Chamber of Princes.

Neither institution was happy with the arrangement, and there was constant pressure on the Court to move out. It wasn't till 1958 that the new Supreme Court premises were built in what was then Hardinge Road and is now Tilak Marg.

THE FIRST ATTACK It is generally believed that the first attack on Parliament took place in 2001. Nair reminds us that it happened on April 8, 1929, when Bhagat Singh and his friend decided to take matters into their own hands.

As she writes "... a deafening noise swallowed the silence. A bomb tore through the floor of the assembly. Smoke rose to fill up the domes of the chamber. For the next few minutes, members ran, some to the nearest door, some to duck behind their desks, a few brave ones leaned in to investigate. Another bomb dropped in quick succession... then came a shower of red leaflets".

There have been two other attacks that were big and bigger. One was in 1966 when a bunch of *sadhus* tried to invade the building over the cow slaughter issue. The other was in 2001 when Pakistani terrorists attacked with guns killing several people.

To conclude, Nair's book is full of interesting nuggets, and one must congratulate her for having dug them all out from the archives. It deserves to be read widely.

The reviewer is a veteran journalist and author

Devastated by Taliban, ignored by all

A heartbreaking account of the fresh wave of cruelty and violence against Afghan women

BOOK REVIEW.

Rasheeda Bhagat

The essence of the book *The Vanishing Girl of Kabul* (Hachette India) by award-winning journalist Zahra Joya and co-authored by Amie Ferris-Rotman is captured in these lines: "The Taliban's oppression was never simply about clothing or schooling. It was about erasing an entire gender from public existence. It was about institutionalising misogyny as a governing principle ... a way to rule through control, fear and the destruction of possibility. Once you normalise misogyny, the rest of the system is easier to break and then control."

This book is yet another heartbreaking account of the fresh wave of cruelty and violence against Afghan women after the US troops walked out of the country in 2021, offering a gift-wrapped nation to the Taliban. Despite their disclaimers at international platforms, including in India, that the Taliban regime would allow girls' education and independence to women, "as allowed under Islamic law", Joya tells us the ground reality is the exact opposite.

TRAGIC TALES Her narrative is sprinkled with horror tales of brutalisation — kidnap/capture, thrashing, torture, forced marriage and rape of innumerable girls and women. Rotman, a veteran British-American journalist and a former *Reuters* correspondent in Kabul, tells us in the introduction that this book is Joya's story. She first met the young girl in 2016 in one of her training sessions in Kabul. The girl had no knowledge of English, but carried a strong passion for reporting. Belonging to the minority

Hazara community of Shias, Joya grew up in Bamiyan, and during Taliban's first regressive regime, attended school dressed as a *bacha posh* (girl who passes for a boy), as her family mourned the birth of girls till the son's arrival. Once they shifted to Kabul, she stealthily worked for a news agency without the knowledge of her parents even while studying. She started her own news agency *Rukshana Media* in 2020, a year before the Taliban returned.

As news came of the Taliban abducting journalists, military interpreters and government employees, many artists, musicians, YouTubers and journalists went into hiding. Joya fled her house, took refuge with friends and relatives, but masked, covered and facing grave danger, continued to send news from Afghanistan to foreign media, including *The Guardian*, which carried her reports.

Within a week it became evident that people like her, "an entire generation of dream-chasers who were now abandoned by the world", would have to flee Afghanistan just to survive. Initially, hundreds of Afghan women marched on the streets protesting as it looked like their rights and freedom were being taken away, but with little success. With help from Rotman, who apprised the British Ambassador in Kabul about the threat to Joya's life, the journalist escaped from Kabul, along with her three sisters, brother and niece, packed with a few hundred, in a British military plane.

The rest of the narrative, written in a simple, engaging and deeply moving style, is about the siblings making London their home. While Joya, the eldest, continues to report on the atrocities and violence against Afghan women, as she has numerous sources back home, her younger siblings get an education. Her reportage is focused on the brutalisation and dehumanisation of Afghan women. Soon recognition and



Title: The Vanishing Girl of Kabul
Authors: Zahra Joya and Amie Ferris-Rotman
Publisher: Hachette India
Price: ₹799

ABOUT THE AUTHORS
Zahra Joya, an Afghan journalist now exiled in London, is the founder of Rukhshana Media, a news outlet covering women's issues in Afghanistan
Amie Ferris-Rotman is the global news editor at New Lines Magazine

awards poured in. In 2022, she visited the US for the first time, to receive an award from the Gates Foundation. Here, she met Nobel Laureate Malala Yusufzai — the Taliban are their common enemy. After her speech, when she was asked if there was any hope for the women of Afghanistan, Joya could barely mask her anger and bitterness at the manner in which the US "had abandoned Afghanistan and moved on. They

seemed to have forgotten that their embassy in Kabul no longer operated, that it was their own officials and soldiers who left it charred and singed," she responded. Joya's seething anger at the shameful role the US regime played throughout the Doha negotiations, where the Afghan government had no role or voice, runs across this narrative, and shows the deep resentment Afghans, particularly its women, harbour against America for being "betrayed".

AUTHENTIC ACCOUNT This book makes compelling, though deeply disturbing, reading, offering a graphic and authentic account of the sexual violence the Taliban continue to inflict on Afghan women. Bahar, who is studying management in Iran, returns home to celebrate Nowruz, but is detained at the border by the Taliban, who thrash, strip and repeatedly rape her till she passes out. She wakes up to find one vertebra fractured and needs three months of treatment.

As Bahar and other rape victims contact Joya to tell their stories, the Taliban were now going into homes, raping mothers and daughters, including little girls. The journalist herself suffers from acute trauma and frequent nightmares. But as she hammers out more such horrendous tales of brutality, more brave women come forward to tell their stories. The international media carries her reports; the atrocities against women are collated and put up before the ICC at The Hague.

As you turn the last page, you can hear the blood-curdling screams of Afghan women imploding the rest of the world: are you simply going to turn your backs on us, leaving us at the mercy of the Taliban?

The reviewer is the editor of Rotary News Trust, and a veteran journalist. She had travelled to Afghanistan in the mid 2000s and reported from there

thehindubusinessline.

TWENTY YEARS AGO TODAY.

September 21, 2006

Fertilizer companies may retain super profits

The Government is planning a new policy initiative to encourage fertiliser production in the country. It is contemplating a move to exempt fertiliser manufacturers producing beyond their stated capacities from sharing their extra gain with the Government.

Indian tourist flow to Thailand unabated

The political uncertainty in Thailand following the coup against the Prime Minister, Mr Thaksin Shinawatra, has not dampened the enthusiasm of Indians eager to visit that country. Leading travel agents as also the two State-owned airlines have reported that it is business as usual as far as Indians travelling to Thailand is concerned.

Markets rally on sustained FII inflows

Equities, currency and bond markets witnessed fresh gains on Wednesday on the back of sustained FII inflows and easing of oil prices. The rupee strengthened beyond Rs 45 a dollar, the first time in the last two months. The benchmark BSE Sensex gained 138 points to touch 12,109. Bonds prices witnessed a single-day gain of 75 paise.

NEW READS.

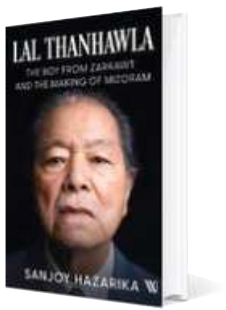


Title: In the Realm of the Last Man

Author: Francis Fukuyama

Publisher: Profile Books

Profound and provocative, this is a study of contemporary politics — and a treatise on how the West can claw democracy back from the brink

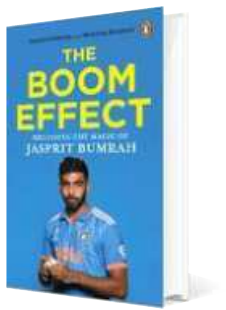


Title: Lal Thanhawla: The Boy from Zarkawt, and the Making of Mizoram

Author: Sanjoy Hazarika

Publisher: Westland Non-Fiction

The book carefully traces Mizoram's rocky road towards peace and stability



Title: The Boom Effect: Decoding the Magic of Jasprit Bumrah

Authors: Aayush Puthran and Nicholas Brookes

Publisher: Penguin Play

The book explores how Bumrah became one of cricket's most entertaining, enigmatic and influential figures

Short take

Food business must look beyond economic gain

Priya Joshi

Recent inspections by Food and Drug Administrations across India have attracted public attention, with spoiled or expired food seized from reputed eateries and establishments found non-compliant in food safety, hygiene and waste management. These are table stakes, the absolute bare minimum for any establishment working with food. Yet compliance, by definition, is the floor. It cannot be the ceiling of our ambition for food.

The real opportunity for any food business is to improve health, strengthen livelihoods, protect biodiversity and celebrate cultures. The question is no longer simply whether a

food business is doing harm. It is whether it can actively create good.

For India, this is particularly consequential. We are one of the world's largest food producers, home to extraordinary agricultural and culinary diversity, while also confronting malnutrition, lifestyle diseases, food waste, water stress and rapidly changing consumption patterns. Our food businesses sit at the centre of many of these contradictions and have an opportunity to become part of their solution.

FAO's Strategic Framework 2022-31 characterises a resilient and sustainable food system through four pillars: better production, better nutrition, better environment and better life. Around the world, institutions and food businesses are beginning to redefine what

responsible food systems can look like. The World Association of Chefs Societies' 'Feed the Planet' initiative promotes sustainable food practices, while the Relais & Châteaux group has made commitments around sustainable farming, biodiversity and protecting local culinary heritage.

The lesson is not that every food business needs to replicate these commitments. It is that ambition can be deliberately designed into the business model rather than added later as a sustainability initiative.

So, what does this philosophy look like when translated into everyday business decisions?

Companies can begin by examining five dimensions: sourcing strength, including transparent supply chains and fair prices for producers; product

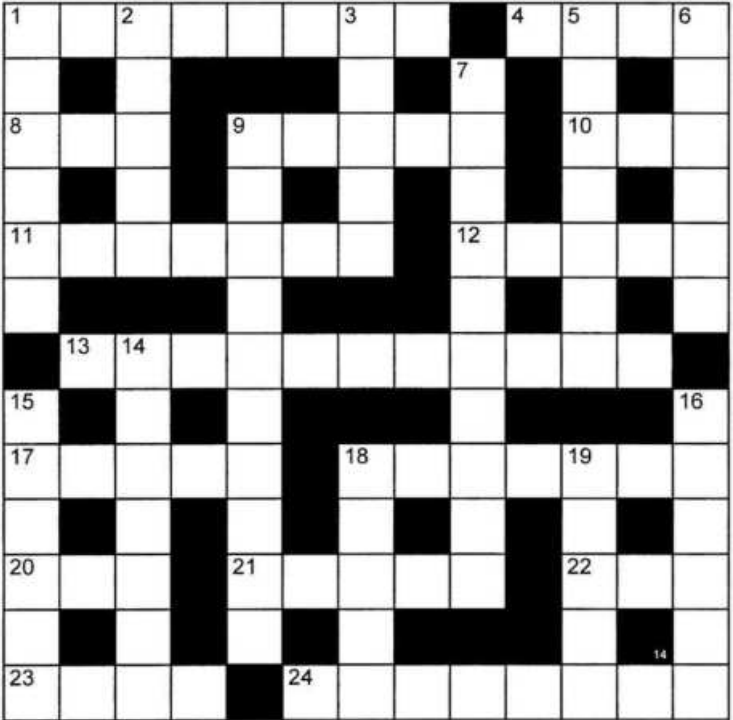
integrity, including nutritional quality, seasonality and ingredients; operational impact, including water, energy, transportation, packaging and waste; people practices, including pay, training and inclusion; and customer experience, including transparency and responsiveness.

The purpose is not to create another sustainability scorecard. It is to help businesses see the connections between decisions often treated as separate.

Some changes can be simple: better food-waste management, shorter supply chains for local ingredients, greater transparency with consumers and sustainability training for teams. Bigger shifts in how products and services are designed can follow over time.

The writer is Co-Founder, Foodwize

BL TWO-WAY CROSSWORD 2739



EASY

ACROSS

- Cloth (8)
- To pack, put under hatches (4)
- No goals (3)
- A simultaneous discharge of artillery (5)
- Tavern, hostelry (3)
- To get in touch with (7)
- A mere show or semblance (5)
- Felt for, or with, another (11) 17.
- Earnestly ask (5)
- Gaming disc (7)
- Young goat (3)
- Female relative (5)
- Employment (3)
- Extinct bird (4)
- Outlook (8)

DOWN

- To threaten (6)
- A hooked claw (5)
- To distribute in portions (5)
- Barely run (liquid) (7)
- Growing less, as the moon (6)
- To give, along with others (10)
- Strike while meeting blow from a part of the stern (7)
- How to tease a child that it's leather (3)
- Pope's daughter unfortunately took five away from Venice (5)
- Habit starts, unless suspicion ends (3)
- One party after another for a bird that's no longer with us (4)
- Respite, cessation of fighting (5)

NOT SO EASY

ACROSS

- I am alert to change that isn't spiritual (8)
- Pack the first seam with fibres of jute (4)
- In the Netherlands one didn't score a goal (3)
- Clause that saves one from an artillery salute (5)
- It's a small hotel such as Lincoln's (3)
- Get in touch with a sort of lens (7) 12.
- The voucher is all right if ten are around (5)
- Felt for another, to empty his sad version of it (11)
- To carry on a lawsuit with some simple advocacy (5)
- Strike while meeting blow from a part of the stern (7)
- How to tease a child that it's leather (3)
- Pope's daughter unfortunately took five away from Venice (5)
- Habit starts, unless suspicion ends (3)
- One party after another for a bird that's no longer with us (4)
- A chance of future success may make one go looking for gold (8)

DOWN

- It threatens fellows with service they can't return (6)
- Hooked claw into the cards left after dealing (5)
- How to allocate everything to the turnover (5)
- Out of kilter with first claret that will barely flow (7)
- Is getting less pale in bronzing finally (6)
- Put in one's bit, on truce negotiation (10)
- Is in a headlong rush to spend it : mag gets it wrong (10)
- How one batted eyelid on day one gave up (7)
- A gun is rendered useless by having drug added (6)
- May be zealous to have such spirits distilled (6)
- How can it make one happier to yell Bravo? (5)
- Cessation of hostilities saw the Engineers cut to pieces (5)

SOLUTION: BL TWO-WAY CROSSWORD 2738

ACROSS 1. Severance 5. Tow 7. Aunt 8. Ignition 10. Nominate 11. City 13. Repeat 15. Will'd 18. Corn 19. Fish-tail 22. Mistaken 23. Long 24. All 25. Hatchment
DOWN 1. Spanner 2. Venom 3. Negate 4. Evil 5. Trivial 6. Windy 9. Knead 12. Fight 14. Parazol 16. Delight 17. Silent 18. Comma 20. Adobe 21. Cash

STURDY COURIER



BonV Aero develops heavy-lift UAV for high-altitude regions

M Ramesh

Bhubaneswar-based drone maker BonV Aero, which is one of the five finalists selected for the GENIUS NY accelerator programme of the State of New York, is developing an ultra heavy payload, high-altitude drone capable of carrying more than 100 kg to altitudes above 10,000 ft and over distances exceeding 100 km.

The capability is particularly relevant to military logistics in India's mountainous and other high-altitude regions. BonV Aero is already a supplier of logistics drones to the Indian defence sector.

Its existing drone can carry a 50-kg payload, fly above 10,000 ft and remain airborne for about 30 minutes, the company's Co-founders and CEOs Satyabrata Satapathy and Gaurav Achha told *businessline*.

Their new drone is designed to tackle challenges specifically involved in operating at high altitudes, where the thinner atmosphere makes it more difficult to generate lift. According to Satapathy, a key differentiator in BonV Aero's drones is their ability to operate in such conditions.

Unlike the carbon-fibre structures commonly used in drones, BonV Aero's aircraft use an aluminium alloy monocoque body. Satapathy said the choice was made taking into account the very low temperatures encountered at high altitudes.

Another feature, according to the company, is the use of multiple rotors that continue to contribute thrust during the cruise phase, rather than becoming redundant after take-off. The forward and rear rotors are operated at different speeds, with the rear rotors being faster, while the propulsion system combines batteries with a fuel-cell system.

BonV Aero says its drones are designed and manufactured entirely in-house, with no imported components. The drones are light enough that just two people are needed to carry them and deploy them anywhere. Satapathy said this enables the company to offer its drones at about one-fifth the cost of comparable Chinese products.

The company is currently building a manufacturing facility in Bhubaneswar at an investment of about ₹300 crore. The plant's planned production capacity is 500 drones a month, compared with the current capacity of about 200 drones a month.

The selection for GENIUS NY gives BonV Aero entry into a year-long, in-residence programme with product and business development support, access to investors and industry leaders, and exposure to the growing uncrewed systems ecosystem in New York.

The five finalists will compete for a total of \$3 million in direct investment — a \$1 million grand prize and four awards of \$500,000 each.

For BonV Aero, the programme also offers a route to the US market for its heavy-lift unmanned aircraft.

GRID STABILITY

The iron-air storage solution for solar

HOME ADVANTAGE. How local minerals can help solve India's solar curtailment problem at a lower cost vis-a-vis Li-ion batteries

K Bharat Kumar

India's accelerated efforts to raise renewable energy capacity in the coming years face at least one significant challenge — lack of mature storage technologies. Renewable power generation firms are often forced to stop production at midday, which is peak hour for solar power production, as there isn't enough demand for such energy.

Theoretically, power can be stored in largescale batteries and used when demand rises in the evenings. But the country does not have such storage capacity due to the high cost of imports and lack of indigenous technology.

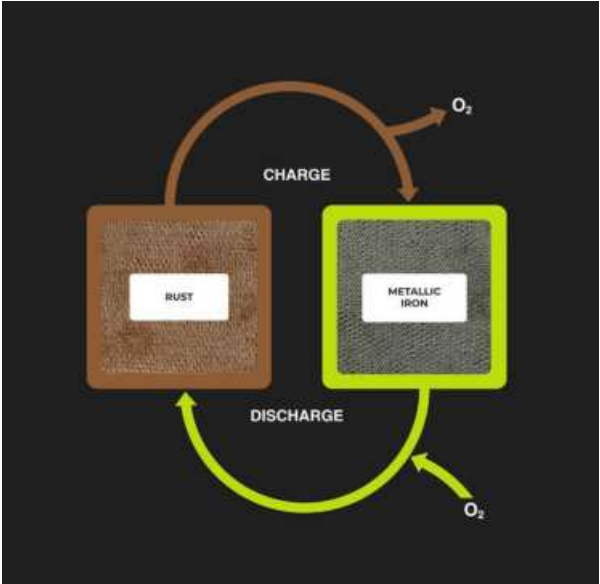
Producers have seen 15–20 per cent average curtailment — namely having to stop production due to lack of demand — with at least one instance of curtailment spiking to 65 per cent last year due to lack of grid-scale storage.

According to Prof Aravind Kumar Chandiran, Faculty, Department of Chemical Engineering, IIT-Madras, “We see from our studies that electricity consumption is projected to triple in India from about 2,000 terawatt hour (TWh) today to about 6,000 TWh by 2047.”

He reasons this would mean about 4 TWh of daily energy storage capacity would be needed, assuming only 50 per cent of power would come from renewables and only half of this would need storage for future use.

As India cannot afford to rely only on imported lithium to meet storage demand of such magnitude, it is important to bring in alternatives, such as domestic mineral solutions.

Lithium-ion batteries offer high power delivery and are quickly responsive. They are ideal for electric mobility and transient, high-power



PROVING METTLE. Iron-air battery workflow; and Priyansh Mohan, Co-founder and CEO of Meine Electric, with the firm's fast-charging iron-air battery system

grid spikes. Metal-air technologies (such as iron-air or zinc-air) offer low power delivery but for a longer duration. These are best suited for steady-state grid storage.

Unlike lithium-ion battery manufacturing, which requires expensive, highly controlled clean-room and ultra-dry environments to prevent chemical instability, iron-air systems can be fabricated in ambient atmospheric conditions, drastically reducing production and capital expenses.

Metal-air systems suffer from what is called 'high electrochemical overpotential' — they need more voltage during a reaction, such as the charging or discharging of a battery. This results in significantly lower efficiency compared to lithium-ion technology. For example, iron-air yields an efficiency of 23–50 per cent, while for zinc-air it is 40–80 per cent, according to Prof Chandiran.

Notably, he mentors this field directly, with three of his former PhD students having founded a

start-up focused on commercialising indigenous zinc-air battery technology for stationary grid storage. Further, lithium-ion battery is a mature technology, whereas metal-air batteries still face chemistry hurdles, engineering scaling challenges and a lack of long-term real-world testing.

LONG-DURATION STORAGE

Startup Meine Electric disclosed in July that it had developed the world's first fast-charging iron-air battery system. This could likely offer a breakthrough in long-duration energy storage.

Iron-air batteries generally operate on a basic reduction-oxidation mechanism — iron reacts with oxygen and water during discharge to form iron oxide, releasing electrical energy. Reversing this reaction forces the iron oxide back into metallic iron.

“Traditional iron-air systems are restricted to multi-day cycles, making them largely incompatible with the intermittent nature of renew-

able energy generation,” says Priyansh Mohan, Co-founder and CEO of Meine Electric. “Solar-heavy grids don't need a 100-hour battery. Unlike regions that require seasonal storage, the renewable challenge across Asia is fundamentally a daily balancing problem.” He says Meine has engineered its system to capture a full charge within a six- to eight-hour window. This has helped compress the multi-day charging window to six hours, paired with 18 hours of discharging.

The company says its technology has been tested and validated by the Customized Energy Solutions (CES) Battery Laboratory. The lab confirmed the “operational stability, capacity retention and electrochemical performance under this 6/18-hour cycle”, Meine stated.

According to Mohan, using Meine's proprietary technology, the storage cost over the battery's lifetime would be about ₹5 per kWh. The clincher here is as much the low cost of operation as the raw material iron, as India has an abund-

ance of iron ore and is the fourth-largest producer globally. However, there is still a long way to go before indigenous and inexpensive technologies see widespread use.

According to a report on the iron-air market released by datamintelligence in August, “The market continues in validation stage. Lack of commercial operational experience, bankability issues, lack of warranty confidence and performance assurance could hinder adoption in the near term.”

Prof Chandiran concurs. “Metal-air technologies will require about five years to prove themselves fit for largescale use. They need to be tested in the real world over thousands of charge-discharge cycles. There are engineering hurdles, scale hurdles, issues with lack of operational experience at scale, all of which we have to surmount.”

We value your feedback.
Do send your comments to
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WOW. In search of a mate



M Ramesh

In the summer of 2025, three fellows who, by all accounts, were desperate left their home in Kafue, Zambia, to look for females to mate with. Their world was shrinking — not many opportunities. Little did they know they were marked. Somebody kept an eye on them. They travelled on foot far into wild, unknown territory. When they encountered cities they skirted them. They came to rivers and walked along the bank until they found an easy point to cross. They walked around large reservoirs. They persevered doggedly.

In April 2026, they reached a place that was 420 km from where they started, and they had walked 2,506 km to get there. And then they walked no more. They appeared to have “settled”.

Their watchers surmise that they found their females.

After all, the Luano Game Management Area in the Luangwa National Park is known to hold a large population of African wild dogs. This remarkable journey has been recorded in the recent issue of *Ecology*, in a paper titled ‘Extraordinary movements connecting distant ecosystems by African wild dogs’. Such journeys have been recorded before — animals go a long way to find mates. The *Smithsonian* magazine, which has a story from the paper, highlights a telling message from the dog episode: The importance of connected landscapes for conservation.

Today, with tagging and GPS, it is possible to take a close peek into the fascinating lives of animals. Jack London could only imagine it in his 1903 adventure novel *The Call of the Wild*. London took his canine hero, Buck, from his cosy home in California to the wilderness of Alaska, where he joined the other descendants of his ancestors — thus truly getting back home.

HEATING UP

Headway in solar thermal

M Ramesh

The Ministry of New and Renewable Energy has given a ‘call for proposals for concentrating solar thermal technology’. Solar thermal — heat rather than electricity from sunlight — is an efficient use of solar energy. Against the backdrop of the MNRE's call, it is useful to look at two solar thermal technologies cooking in the labs.

One, according to details published in *Energy*, is about combining low-temperature solar thermal collectors and high-temperature heat pumps, using ethanol as the working fluid (refrigerant).

Industries often need heat at 120–200 degrees C. Solar thermal systems typically deliver much less. However, by combining with heat pumps, researchers at the University of Guanajuato and Texas A&M achieved 155.6 degrees C.

The process is simple. Step 1 is to use solar collectors to heat water to about 99 degrees C. The hot water is stored in an insulated tank. Step 2 is to use the hot water to heat the pump's evaporator, turning the refrigerant (ethanol) into vapour. The third step is to raise the refrigerant's temperature by compressing the vapour.



BOOSTER. A heat pump can act as a thermal elevator for solar collectors ISTOCK

In the industrial process under study, the refrigerant reached about 156 degrees C — enough heat for making tomato paste — before condensing and returning through an expansion valve for the next cycle.

The heat pump effectively acts as a thermal elevator: Solar energy enters at a low temperature, and electricity drives the compressor to deliver useful heat at a considerably higher temperature.

The researchers tested four working fluids. Ethanol emerged the strongest performer. At a temperature lift of 70 degrees C, the heat pump achieved a coefficient of performance (COP) of 4.64, which means that every unit of compressor electricity was associated with 4.64 units of heat delivered. The calculated ‘exergy efficiency’ was 74.4 per cent, while the levelised cost of energy was \$0.0818 per kWh.

The result is significant because the solar collectors themselves do not have to operate at the process temperature. Instead, relatively inexpensive low-temperature solar heat is upgraded electrically.

The authors estimate that the system could avoid about 3,613 tonnes of carbon dioxide a year, and the ethanol-based configuration had a simple payback period of about 8.5 years in the case studied.

The second study, published in the *Journal of Energy Storage*, is about making an age-old idea — compressed air storage — work.

You can store energy by compressing air using electricity when available cheaply. But the problem arises during the discharge cycle, when the air is let out to drive turbines. When let out, the compressed air decompresses and becomes cold, leading to problems such as ice formation on turbine blades. However, more importantly, cold air lacks energy and cannot turn turbines.

A group of researchers from Spain, Italy and Switzerland have tackled this problem with a ridiculously simple trick: Use solar thermal energy to reheat the compressed air before it enters the turbines. The researchers modelled this in Chile. They used electricity to compress air into an underground cavern, while solar energy heats air and this goes into a thermal energy storage system. During discharge, the stored heat is used to reheat the compressed air before it expands through the turbines.

There is another useful twist. The heat released during the compression can be recovered and supplied to a nearby industrial user, thereby improving the economics of the system. The researchers envisage industrial heat of 140–160 degrees C.

The economic viability of the two systems depends on factors such as the cost of input electricity and the prices that the output electricity and recovered heat will fetch. But they clearly show a pathway that can be improved.

Neither technology is yet a universal competitor to conventional systems. The ‘compressed air energy storage with concentrated solar power’ (CAES-CSP) study, for instance, puts the concept only at technology readiness level (TRL) 3–4, compared with TRL 9 for battery energy storage systems and stresses that its economics depend on the unusual coincidence of strong solar resources, suitable geology and a nearby industrial heat consumer.

But that may be precisely where the opportunity lies: Instead of asking how much electricity a solar plant can produce, ask what combination of electricity, heat and storage it can deliver.

When solar glass becomes a power generator

US company Caelux achieves this by coating the glass with a thin film of perovskites

M Ramesh

Solar cell and module manufacturer GREW Solar recently announced a five-year commercialisation partnership with US-headquartered Caelux Corporation to produce high-power hybrid-tandem solar modules using Caelux's ‘energy producing glass’. Here, the glass covering the solar cells itself is an energy producing layer.

Caelux achieves this by coating the glass with a thin film of perovskites. Vish Iyer, Managing Director–India and West Asia, says the perovskite layer captures sunlight that the silicon cell cannot use as efficiently, while the cell itself continues to perform as before.

The solar industry has focused on two approaches to perovskite technology: Integrating perovskite into the silicon solar cell and developing perovskite as an energy pro-

ducing layer. Caelux's R&D focused on the latter, Iyer tells *businessline*.

The distinction is important. In the conventional perovskite-silicon tandem approach, the perovskite is deposited directly on the silicon cell, requiring changes to cell architecture and perhaps new manufacturing equipment. Caelux, instead, deposits the perovskite solar layer on the glass that normally serves as the module's front cover.

“In effect, we turn that inert glass into an energy producing layer,” Iyer says.

This allows the use of existing silicon module factories and equipment. Caelux says its energy producing glass can be paired with mono-PERC, TOPCon, back contact and heterojunction technologies, and with modules 1.7–2.5 m long.

Rather than replacing silicon, it is about adding another generator to the same module.



SURFACE POWER. Caelux solar modules with ‘energy producing glass’; and Vish Iyer, Managing Director–India and West Asia, Caelux



For instance, a 10 GW TOPCon manufacturing line could, according to Caelux, produce the equivalent of 12 GW of module output after incorporating its technology with limited additional investment.

The higher power density could also reduce the requirement of land, mounting structures, cabling and other components.

Caelux has signed agreements with Indian manufacturers, including Rayzon Solar and Navitas Solar, for 5

area,” Vineet Mittal, Director-Finance and Strategy, Navitas Solar, says. “If successfully commercialised at scale, moving module efficiencies from around 23 per cent today towards 28 per cent and beyond could strengthen India's ability to manufacture next-generation, high-efficiency solar modules for both domestic and international markets.”

Perovskite is a thin-film technology, and the challenge is in depositing a uniform layer over a large area and ensuring it survives decades of outdoor exposure.

Caelux says years of R&D have gone into solving these issues. The company shipped the first commercial order of its active glass technology in 2025.

Its medium-term plan includes production in India, alongside its US operations.

The next generation of solar modules may not require abandoning the old one but simply making the glass work.

MY FIVE.

SAURABH AGARWAL
Director and CEO, HAB
Pharmaceuticals and
Research Ltd



Run on Marine Drive, Hyrox training and blue lights out

- 1 Marine Drive run: My day usually starts at 5 am. Two glasses of warm water, a banana, and I'm out for a run on Marine Drive or a gym session. The sea at that hour is the calmest meeting I'll have all day.
- 2 Hybrid athlete: I'm into hybrid fitness, a balance of endurance and strength. I eat for both. Breakfast is fruit, eggs and a smoothie. Lunch and dinner are balanced plates of carbs, protein and fibre.
- 3 Training partner: Two days of running, two for Hyrox training, one for weights and one rest day. My wife Gayatri and I race Hyrox doubles together, so fitness becomes a shared goal. The rest day is non-negotiable.
- 4 Serious recovery: Intense training needs serious recovery. I track my recovery score, HRV, resting heart rate and VO2 max on Whoop — to know when to push and when to back off. NormaTec compression, deep tissue massages and cryotherapy are regulars in my week.
- 5 Blue lights out: Thirty minutes before bed, screens go off, blue light blocking glasses go on to cut the glare from indoor lighting. Am in bed before 10, because 5 am arrives quickly.

CAPSULE.

China, South Africa tie-up

The Southern African Development Community (SADC), UNAIDS, the Chinese government and the Global Health



Innovation Institute (GHI) have agreed to advance collaboration on regional pharmaceutical manufacturing, following a joint visit to production facilities in

Johannesburg, Durban and Cape Town in South Africa, a note from UNAIDS said.

The delegation assessed existing capabilities in pharmaceutical and vaccine manufacturing, biotechnology, genomics and diagnostics, and identified priority areas to help Southern African countries expand the domestic production of quality-assured medicines and health technologies.

The collaboration aims to reduce reliance on imports, strengthen health security and improve timely access to essential health products across the region. Partnering with China offers an opportunity to connect technology, expertise, investment and manufacturing capabilities with this growing regional capacity, it added.

The collaboration aligns with the United Nations' 'Political Declaration on HIV/AIDS,' which calls for greater health sovereignty and strengthened cooperation to advance local and regional production of HIV health technologies as part of efforts to end AIDS as a public health threat by 2030. It also comes at an important moment following the UN Day for South-South Cooperation.

COMING UP.

Joining hands to beat rabies

The theme for World Rabies Day (September 28) this year is 'Stronger together', highlighting the fact that rabies elimination is possible only when we work together. "Rabies elimination depends on connected systems. Stronger partnerships create stronger outcomes," said the World Health Organization, adding that rabies is a vaccine-preventable, zoonotic, viral disease.

CARE WITH CONSENT

‘Break-glass’ allowance: How safe is personal health data?

DOS AND DON'TS. Medical bodies can now access patient data in exceptional cases, but with caveats

Vallari Sanzgiri

Consider a situation that confronts medical practitioners, more often than not, in a hospital emergency room — an unconscious, unaccompanied patient is wheeled in, who may possibly require insulin, for example.

Unlike in the past, health professionals and medical institutions can now access patient data in such exceptional situations under the Digital Personal Data Protection Act. But this special “break-glass” allowance comes with caveats, including informing the patient of this forced access, otherwise known as a ‘breach notification’. In fact, this provision covers all health data, including from diagnostic labs, where one’s blood and other granular data is collected, besides specialist interactions where a patient may not comprehend how their health details may be processed.

Essentially, the DPDP Act mandates every entity processing personal information to explain how they are processing patient information, what is included in the patient information, how long this information will remain with the entity, and who is allowed to access this information. Failure to disclose these details, enforce data safeguards or even inform a patient of a data breach makes the health entity liable to a whopping fine of ₹250 crore.

The Act does come as a huge relief to patients across India — since personal health data is coveted by insurance companies, private entities and, at times, even employers who may alter their employment contracts based on the employee’s health condition. What is left wanting are granular rules that further protect patients, even as they help healthcare institutions un-



ISTOCK

derstand data protection and governance in various use-cases.

“Patient privacy moves from something healthcare institutions cared about to something they will increasingly have to demonstrate. Health data has always been among the most sensitive information a healthcare provider holds. The sector has long operated under obligations spread across medical regulations, sector-specific laws and frameworks. DPDP changes the accountability layer,” explains Malcolm Gomes, Chief Operating Officer, IDfy, and Head of Privy by IDfy (which assists enterprises with DPDP compliance).

CONSENT MECHANISMS

Despite the consent-based approach, health data under the DPDP Act does not carry a separate label of ‘sensitive personal data’. Such a categorisation would have ensured additional privacy and safeguards such as “opt-in” consent mechanisms, explicit prohibition on using the data for employment purposes,

and pushing healthcare entities to undertake data protection impact assessment surveys. Further, the label would have placed restrictions on continuous collection of health data.

By clubbing health data with general personal information, the Act misses offering additional safeguards against potential misuse in the form of discrimination, stigma or loss of autonomy, says Shivangi Rai, Deputy Coordinator and Lead (technology, health and society), Centre for Health Equity, Law and Policy.

DATA MOVEMENT

Beyond categorisation, health institutions also need to understand how to safeguard patient data that moves across the system. A patient is bound to approach multiple doctors, clinics and hospitals when seeking treatment. The Act does not explain how these separate entities should coordinate with each other. Healthcare organisations also need to understand how the Act will work alongside existing clin-

ical, insurance, record retention and digital health requirements.

“Questions around retention, emergency use, research and secondary use of health data will, therefore, require careful governance rather than a one-size-fits-all approach. Children’s data introduces another layer of complexity, particularly around verifiable parental consent,” says Gomes.

FOREIGNERS' TREATMENT

The entry of foreign patients further complicates these processes. A foreign patient will have to share old treatment records and insurance details with an Indian hospital before travelling for surgery. Post surgery, the hospital must share treatment updates, discharge summaries and billing records with the insurance company.

While the current law allows this, it also mentions “no transfer” or “restricted” countries, where information cannot travel.

“As of September 2026, no country has been notified as a restricted destination. However, the provision is part of the DPDP framework, whose substantive implementation is being phased in. Organisations should not treat the absence of a notified blacklist as the end of their cross-border compliance analysis,” says Rishi Agrawal, CEO and co-founder, Teamlease Regtech.

India’s medical tourism industry is projected to reach \$16.21 billion by 2030 at a CAGR of 13.23 per cent, drawing over 4.5 lakh foreign patients annually, according to a Deloitte-Confederation of Indian Industry report.

Agrawal also differentiates between the law being enacted and enforced.

Notified in 2023, the Act and its related rules are being enforced in a phased manner. The watchdog (Data Protection Board of India) has also been established. In eight months, the law and its penalties will kick in.

FROM THE REGULATOR

Blueprint for AI use in the healthcare sector

The UK’s National Commission into the Regulation of AI in Healthcare has published its recommendations on how that country can regulate AI in healthcare to ensure it remains safe, keeps pace with innovation and continues to earn people’s trust.

AI is already used in the NHS to spot, early on, serious health problems such as strokes and skin cancer, and free up clinicians’ time through voice-enabled technologies. The commission’s recommendations aim to help more patients benefit safely and quickly, by supporting the UK to become the best place for innovators to build and safely test AI, the best place for healthcare professionals to use it and, most importantly, the best place for patients to engage with it in their care.

The independent commission was established by the Medicines and Healthcare products Regulatory Agency (MHRA) in September 2025 and includes diverse expertise across health, regulation and AI, including international contributors from the US and Singapore.

It gathered evidence over the course of a year from more than 12,000 people, including patients, the public, clinicians, healthcare leaders, industry and technology developers, making it the largest engagement of its kind in the UK on the regulation of healthcare technology. Additional research by the Health Foundation provided detailed insight into the UK public’s priorities and expectations for the regulation of AI in healthcare.

Taken together, the engagement found broad public support for AI in healthcare, with clear conditions: Strong safety standards, meaningful human oversight and transparency about when and how AI is being used in treatments. The commission’s recommendations respond directly to these views.

It recommends that the MHRA introduce staged authorisations for new AI models, similar to ‘L-plates’ for learner drivers. New models would be deployed under close supervision and with tight guardrails, allowing them to demonstrate real-world safety and performance before being granted fuller authorisation. This staged approach is designed to give UK patients world-first access to promising new models, while ensuring risk is carefully controlled at every step.

Source: MHRA



ISTOCK

Missing link in Ayushman Bharat: Emergency care

BY INVITATION



K MADAN GOPAL
KS UPLABDH GOPAL



GOLDEN HOUR. Timely, trained care saves lives and limbs ISTOCK

Eight years after its launch, the Ayushman Bharat scheme has shown that India can create health entitlements at an unprecedented scale. More than 45 crore Ayushman cards have been issued, and 12 crore hospital admissions authorised.

The scheme has developed a wider health system architecture: The Ayushman Arogya Mandirs strengthen primary care, PM-ABHIM is adding critical-care infrastructure, and the Ayushman Bharat Digital Mission (ABDM) is building the digital backbone. The challenge is in making these building blocks work together. Nothing illustrates this gap more than a medical emergency.

A person with a serious road accident injury, heart attack, stroke, sepsis or obstetric emergency does not enter the health system in an orderly manner. What matters is a quick response, trained care before hospitalisation and timely access to appropriate treatment.

India is not starting from scratch. The 112 Emergency Response Support System provides a national emergency number, while States operate ambulance services under the National Health Mission. The National Ambulance Service Operational Guidelines, 2026, aim for standardisation in despatch, triage, tracking and emergency medical technician (EMT) skills.

The recently introduced PM-RAHAT scheme for road accident victims provides an important facility. It enables cashless treatment of up to ₹1.5 lakh for up to seven days at designated hospitals. When life is at risk, treatment should not wait while a family arranges money or establishes insurance entitlement.

An ambulance may reach a patient without knowing which nearby hospital has the required specialist, blood, operating theatre or intensive-care bed. A patient may be taken to the nearest hospital only to be referred again, with little information

about what happened during transport. Critical care blocks have limited value if the referral system cannot bring in the right patient in time. For families, these are lost minutes.

The 112 platform and State ambulance systems could form the backbone of an integrated emergency response, with command centres tracking ambulances and hospital capacity. A severely injured person should be taken not simply to the nearest hospital but one capable of managing that injury. Ambulances must become an extension of clinical care, with trained EMTs.

At the hospital, a critically ill patient should receive immediate stabilisation without waiting for a deposit, identity document or insurance approval. A transparent financing mechanism could enable participating private hospitals to provide initial stabilisation without shifting the cost to families. PM-RAHAT provides a useful pathway.

District and select sub-district hospitals should also form part of a graded emergency care network with defined capabilities and referral pathways. ABDM can help create this continuum by supporting pre-arrival information, and referral and transfer summaries with privacy and consent safeguards.

The journey also begins and ends in primary care. Improved control of hypertension and diabetes can, in fact, prevent some emergencies, while rehabilitation and follow-up must continue after discharge.

Dr Madan Gopal is a former senior health consultant with NITI Aayog, and Dr KS Uplabdh Gopal is a health professional. Views are personal

Wellness-centric care with nutraceuticals

Deciphering the right dose of alternatives to illness-centric, drugs-based healthcare

DB Anantha Narayana

A neighbour walked into my house with a bottle of melatonin tablets, labelled “nutraceuticals”, and wanted to know if he could give it to a relative who had just flown in from abroad.

I gave my “go ahead” as melatonin is not a drug but a hormone produced in the pineal gland of the human brain for sleep and sleep cycle management. It is also manufactured in factories now as a nutraceutical, made available as tablets for convenience, and helps fight jet lag and promote good sleep.

HISTORY OF USAGE

Dr Stephen DE Felice, in 1989, coined the term “nutraceutical” by combining “nutrition” and “pharmaceuticals”. And unlike melatonin, a number of other compounds useful for



LEAFY CURE. Most nutraceuticals have their origin in herbs ISTOCK

health are sourced from herbs.

In fact, herbs have a long history of human consumption (HAHC). In the Indian context, many herbs have a documented long history of safe usage (HoSU) in traditional

knowledge texts. These herbs are processed for consumption through steps such as infusion and decoction, and taken together with milk and so on.

In the last few decades, the industry has converted such

herbal extracts into powders through spray drying and other technologies. They are then sold as powders, granules, capsules, tablets, gels or value-added food stuff, with specified usage for convenience.

Such products help consumers move from illness-centric healthcare through drugs to wellness-centric healthcare.

REGULATORY OVERSIGHT

The products are generally called food or health supplements and are regulated under the FSSAI’s ‘Food Safety and Standards Act and Rules’.

A warm cup of tea is often consumed to sleep better. L-theanine in tea was seen, in human studies, to reduce the lag-time in falling asleep.

A 200 mg L-theanine product in the evening is useful to sleep better. Curcuminoids from turmeric; zeaxanthin and lutein from marigold petals;

resveratrol from the skin of grapes; hesperidin and quercetin from lemon peel are a few examples of nutraceuticals.

Herbs have compounds in them, which are the basis for their health benefits. Scientists have isolated and produced many such compounds from herbs in their pure form. Most nutraceuticals have their origins in one or more herbs.

Scientist Dr Vishweshwaraiah Prakash, former Director of the Central Food Technology Research Institute, Mysuru, coined the term “herbaceuticals” to promote the appropriate use of herbs and compounds for their health benefits. Their consumption, with the right dosage information, is, in fact, the best prescription.

The writer is a regulatory expert, pharmacist and former Chairman, Expert Committee on Nutraceuticals, FSSAI. Views are personal

